PHOTOGRAPHS BY WALTER HALL

A home-built vacuum chamber

Walter Hall makes his own vacuum chamber for stabilising pen blanks

WALTER HALL

Walter Hall is a woodturner who has specialised in making pens and pencils for more than 20 years. Based on the beautiful Northumberland coast in the UK, Walter sells his bespoke pens and pencils through local craft centres and via his website.

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EQUIPMENT USED
Pressure cooker
Vacuum supply fittings: vacuum gauge, valves, four-way connector, straight connector, adaptor
20mm-thick sheet of Perspex
Pair of dividers
Pillar drill with 3.5mm drill bit
Bandsaw with a narrow blade
3.5mm woodscrew
Abrasives
Polishing compounds
11.8mm drill
¼in BSP tap in a tap wrench
PTFE tape

MATERIALS
Sheet of plywood for a jig
Pen blanks for testing
Resin

Before I begin to explain how I made my own vacuum pot, for those who are unfamiliar with the concept of stabilising wood or perhaps even why we might want to do so, a brief explanation is in order. Often as pen makers or woodturners we may come across an attractively figured or spalted piece of wood that would make a beautiful pen or other turned item but the wood is so soft and punky that it would be difficult or perhaps even impossible to turn and finish. Stabilising the wood by impregnating it with resin hardens it up so that it can be turned successfully.

A number of different chemicals are available for this purpose, some of which need to be mixed with a catalyst before use and heat treated after vacuuming while others come ready mixed and require only a drying out period to prepare them for use. It is beyond the scope of this article to give detailed instructions for the correct use of each type of resin, but comprehensive instructions are supplied with the products and further extensive advice is often available on suppliers' websites.

To test my vacuum pot I used Drechseln und Mehr Stabi 17 B Vakuumharz (Vacuum resin), which does not require heat treatment after vacuuming. Full instructions for use are available on the website – www.drechselnundmehr.de – but you will need to translate them from German using an online translator such as Google Translate.

I had been considering stabilising my own blanks for some time but had been deterred by the high setup costs of a pressure chamber and a suitable vacuum generator or pump so decided to reduce the costs by building my own chamber based upon a retired domestic pressure cooker. This will be large enough for me to stabilise several pen blanks at a time and even to stabilise small bowl blanks. The vacuum is provided by a HoldFast vacuum generator intended for vacuum chucks but will eventually be replaced by a vacuum pump, which will enable a more complete vacuum to be achieved. Useful advice on the selection of a vacuum pump for stabilising is available on the Turntex Woodworks website – www.turntex.com – who are the suppliers of Cactus Juice stabilising resin.

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